

Student Study Tracker System Draft

Team Lovelyy (Group 6)

Sahil Khan Miya, Aditya Yadav, Sujal Lama, Samrat Jung Shahi

Westcliff University

PRG 200

Professor Pandey

June 9, 2026

Student Study Tracker System Draft

This report is about the Student Study Tracker System, which is a Python project being built for the PRG 200 course. The goal of this project is to make a simple system that helps students keep track of how much they study. The app lets users add subjects, record study sessions, set a daily study goal, and see their progress in charts. This report covers the work done in Week 5 and Week 6, the tools used, the problems we ran into, and what is left to do in Week 7 and Week 8.

Technologies and Tools Used

Below are the tools and technologies the team have used to build this project. Some tools will be used in the upcoming weeks which are week 7 and 8 to improve our project.

- Python – The main language used to write all the code.
- SQLite (sqlite3 library) – Used to save and read data from a database file.
- Tkinter – A Python library used to make the app window and buttons.
- Matplotlib – Used to make bar charts and pie charts from the study data.
- Visual Studio Code – The code editor used to write and run all the files.
- GitHub – Used to save and share the project code each week.

Week 5: Basic Project Structure

In Week 5, the team started building the basic parts of the project. A file called week5.py was created to show that the main features of the app were working. At this point, the data was stored in Python lists, not in a database. This meant that every time the program was closed, all the data was gone and nothing was saved.

The features that were built in Week 5 are listed below:

- `add_subject(name)` – Adds a subject name to a list.
- `log_session(subject, duration, notes)` – Saves a study session as a dictionary inside a list.

- `show_subjects()` – Prints all the subjects that were added.
- `show_sessions()` – Prints all the study sessions that were logged.

The problem with using lists is that data is not saved anywhere. Once the program stops running, everything is lost. This showed the team that a proper database was needed, which was the main task for Week 6.

Week 6: Adding the Database

In Week 6, the team replaced the lists with a real SQLite database. A file called `database.py` was made to hold all the database functions. This file is now the main backbone of the whole project. The team also went back and updated the Week 5 code to use the new database instead of lists. This updated version was saved as `week6.py` to show how the project improved from Week 5 to Week 6.

The features added in Week 6 are listed below:

- `init_db()` – Creates the database file and three tables: subjects, sessions, and goals.
- `add_subject(name)` – Saves a subject name into the subjects table.
- `log_session(subject, duration, notes)` – Saves a study session into the sessions table.

Today's date is saved automatically.

- `set_goal(minutes)` – Saves the daily study goal in minutes into the goals table.
- `get_goal()` – Gets the current daily goal from the database.
- `get_today_total(today)` – Adds up all the minutes studied on a given date.
- `get_streak()` – Counts how many days in a row the user has studied.

Problems Found

Problem 1: Same Subjects Added Again and Again

When week6.py is run more than once, it adds the same subjects to the database every time. So, if the file is run three times, the same three subjects get added three times, which gives nine entries instead of three. This happens because the code does not check if a subject already exists before adding it. For now, the fix is to delete the database file called study_tracker.db before running the file again. This will be fixed properly in the final version of the app.

Problem 2: The Demo Files and the Final App Use Different Database Setups

The demo files, week5.py and week6.py, save the subject name as plain text inside the sessions table. But the final app file, main.py, saves a subject ID number instead, which points to the subjects table. Because of this difference, both cannot use the same database file at the same time. If they do, errors will happen. Right now, the fix is to keep the demo files and the final app in separate folders so they each have their own database. This problem will go away on its own once everything is merged into one single folder at the end of the project, because then there will only be one database setup being used.

What is Left to Do

Week 7: Building the App Window

In Week 7, the team will build the full app window using Tkinter. All the features from Week 5 and Week 6 will be connected to buttons and screens that the user can click on. The app will have a main dashboard that shows today's study time, the daily goal, and the current streak. There will also be separate screens for adding subjects, logging sessions, setting goals, and viewing the history of past sessions. The GitHub page will be updated with the new code.

Week 8: Charts, Testing, and Final Submission

In Week 8, the team will add charts to the app using Matplotlib. There will be a bar chart to show how many minutes were studied each day, and a pie chart to show how study time was split across different subjects. The team will also test the whole app to find and fix any bugs. After testing, the final documentation will be written, the presentation slides will be made, and the project will be submitted. The GitHub page will get one last update as well.

Conclusion

The project has come a long way in just two weeks. In Week 5, the team built the basic parts of the app using simple Python lists. In Week 6, a proper database was added so data is now saved permanently. The updated Week 5 code was also redone to use the new database. Two problems were found during this process and simple fixes were put in place for now. By the end of Week 8, the full app with a working interface and charts will be ready to submit. The team feels good about where the project stands and is ready to finish it in the coming weeks.